

Apache Flink Demo Report

Demo này minh họa một pipeline **real-time transaction analytics**:

```
Transaction Producer
-> Kafka topic: transactions
-> Apache Flink Job
-> Parse / Filter
-> keyBy(account_id)
-> 10-second Tumbling Window
-> count + total + average
-> Console Output / Alert
```

Mục tiêu khi quay demo là cho người xem thấy:

- Flink đọc stream liên tục từ Kafka.
- Flink biến job thành dataflow graph trên Web UI.
- Job chạy song song với parallelism/subtasks.
- Output cập nhật theo window.
- Checkpoint hoàn tất để minh họa fault tolerance.

Lưu ý: phần cài đặt/chạy lệnh trong file này dùng để chuẩn bị và quay video. Khi thuyết trình, không trình bày chi tiết các bước cài đặt.

1. Kiểm Tra Môi Trường

Mở PowerShell và kiểm tra:

```
java -version
docker --version
docker compose version
mvn -v
```

Kỳ vọng:

- Java/JDK đã có.
- Docker Desktop chạy được.
- Docker Compose hoạt động.
- Maven hoạt động.

```
PS C:\Users\dnthi> java -version
java version "21.0.8" 2025-07-15 LTS
Java(TM) SE Runtime Environment (build 21.0.8+12-LTS-250)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.8+12-LTS-250, mixed mode, sharing)
PS C:\Users\dnthi> docker --version
Docker version 29.5.2, build 79eb04c
PS C:\Users\dnthi> docker compose version
Docker Compose version v5.1.4
PS C:\Users\dnthi> mvn -v
Apache Maven 3.9.10 (5f519b97e944483d878815739f519b2eade0a91d)
Maven home: C:\Users\dnthi\tools\apache-maven-3.9.10
Java version: 21.0.8, vendor: Oracle Corporation, runtime: C:\Program Files\Java\jdk-21
Default locale: en_US, platform encoding: UTF-8
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
PS C:\Users\dnthi> |
```

2. Cấu Trúc Project

Thư mục demo hiện tại:

```
flink-transaction-demo
|
|-- docker-compose.yml
|-- pom.xml
|-- src
|   |-- main
|   |   |-- java
|   |   |   |-- com
|   |   |   |   |-- seminar
|   |   |   |   |-- flink
|   |   |   |       |-- Transaction.java
|   |   |   |       |-- TransactionProducer.java
|   |   |   |-- TransactionAnalyticsJob.java
|-- target
```

Vai trò các file chính:

File	Vai trò
<code>docker-compose.yml</code>	Dựng Kafka, Flink JobManager và Flink TaskManager.
<code>pom.xml</code>	Khai báo dependency Flink, Kafka connector, Jackson và build fat JAR.
<code>Transaction.java</code>	Model dữ liệu giao dịch để parse JSON.
<code>TransactionProducer.java</code>	Sinh transaction liên tục và gửi vào Kafka topic <code>transactions</code> .
<code>TransactionAnalyticsJob.java</code>	Flink job chính: Kafka source -> parse/filter -> keyBy -> window -> aggregate -> output.

```

Topic: transactions Partition: 1 Leader: 1 Replicas: 1 ISR: 1 ECR: LastKnownEpoch:
PS C:\Users\dnthi\OneDrive\Documents\BigData\Seminar\Theory\demo\flink-transaction-demo> mkdir src\main\java\com\seminar\flink

Directory:
C:\Users\dnthi\OneDrive\Documents\BigData\Seminar\Theory\demo\flink-transaction-demo\src\main\java\com\seminar

Mode                LastWriteTime         Length Name
----                -
d-----            8/15/2026 12:27 AM                flink

PS C:\Users\dnthi\OneDrive\Documents\BigData\Seminar\Theory\demo\flink-transaction-demo> tree /F
Folder PATH listing for volume Windows
Volume serial number is 0000005C 1478:27C0
C:.
├── docker-compose.yml
├── src
│   └── main
│       ├── java
│       │   ├── com
│       │   │   ├── seminar
│       │   │   └── flink

```

3. Docker Compose: Kafka + Flink

File `docker-compose.yml` dựng 3 service:

- `kafka`: Kafka broker.
- `jobmanager`: Flink JobManager, mở Web UI ở `http://localhost:8081`.
- `taskmanager`: Flink TaskManager, cấu hình `taskmanager.numberOfTaskSlots: 2`.

Kafka dùng 2 listener:

- `localhost:9092`: cho producer chạy từ máy host.
- `kafka:29092`: cho Flink job chạy trong container kết nối tới Kafka.

Chạy cluster:

```

cd "C:\Users\dnthi\OneDrive\Documents\BigData\Seminar\Theory\demo\flink-transaction-demo"
docker compose up -d

```

Kiểm tra container:

```
docker ps
```

Cần thấy:

```

kafka
flink-jobmanager
flink-taskmanager

```

```

PS C:\Users\dnthi\OneDrive\Documents\BigData\Seminar\Theory\demo\flink-transaction-demo> docker compose up -d
[+] up 30/30
✔Image flink:2.1-java17          Pulled          70.4s
✔Image apache/kafka:3.9.1       Pulled          29.2s
✔Network flink-transaction-demo_default Created         0.1s
✔Container kafka                 Started         2.3s
✔Container flink-jobmanager      Started         2.3s
✔Container flink-taskmanager     Started         1.4s

What's next:
  Filter, search, and stream logs from all your Compose services
  in one place with Docker Desktop's Logs view. docker-desktop://dashboard/logs?appId=flink-transaction-demo
PS C:\Users\dnthi\OneDrive\Documents\BigData\Seminar\Theory\demo\flink-transaction-demo> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
42790e5b5d2e   flink:2.1-java17                    "/docker-entrypoint..." 3 minutes ago  Up 3 minutes  6123/tcp, 8081/tcp
9c225109c4b0   apache/kafka:3.9.1                  "/__cacert_entrypoint..." 3 minutes ago  Up 3 minutes  0.0.0.0:9092->9092/tcp, [::]:9092->9092/tcp
fe35f7c51bbe   flink:2.1-java17                    "/docker-entrypoint..." 3 minutes ago  Up 3 minutes  0.0.0.0:8081->8081/tcp, [::]:8081->8081/tcp
PS C:\Users\dnthi\OneDrive\Documents\BigData\Seminar\Theory\demo\flink-transaction-demo>

```

Mở Flink Web UI:

`http://localhost:8081`

The screenshot shows the Apache Flink Dashboard interface. The top navigation bar includes the Apache Flink logo and the text 'Apache Flink Dashboard'. The main content area is divided into several sections: 'Available Task Slots' showing a value of 2, 'Running Jobs' showing a value of 0, and 'Completed Job List' showing an empty table. The left sidebar contains a list of navigation items: Overview, Jobs, Task Managers, and Job Manager. The top right corner displays the dashboard version as 2.1.3 and the commit hash as 6cda56b @ 2026-06-08T04:56:20+02:00.

4. Tạo Kafka Topic

Tạo topic `transactions` với 2 partitions:

```

docker exec kafka /opt/kafka/bin/kafka-topics.sh --create --topic transactions --
bootstrap-server localhost:9092 --partitions 2 --replication-factor 1

```

Nếu thành công:

Created topic transactions.

Kiểm tra topic:

```
docker exec kafka /opt/kafka/bin/kafka-topics.sh --describe --topic transactions -
--bootstrap-server localhost:9092
```

Kỳ vọng:

```
Topic: transactions
PartitionCount: 2
ReplicationFactor: 1
```

```
PS C:\Users\dnthi\OneDrive\Documents\BigData\Seminar\Theory\demo\flink-transaction-demo> docker exec kafka /opt/kafka/bin/kafka-topics.sh --create --topic transactions --bootstrap-server localhost:9092 --partitions 2 --replication-factor 1
Created topic transactions.
PS C:\Users\dnthi\OneDrive\Documents\BigData\Seminar\Theory\demo\flink-transaction-demo> docker exec kafka /opt/kafka/bin/kafka-topics.sh --describe --topic transactions --bootstrap-server localhost:9092
Topic: transactions    TopicId: IsBVn0Q6Rvu2GvCOIq1LjQ PartitionCount: 2    ReplicationFactor: 1    Configs:
  Topic: transactions Partition: 0    Leader: 1    Replicas: 1    Isr: 1 Elr:    LastKnownElr:
  Topic: transactions Partition: 1    Leader: 1    Replicas: 1    Isr: 1 Elr:    LastKnownElr:
```

Vì topic có 2 partitions, demo sẽ dễ nối với phần **parallelism/subtasks** hơn.

```
transactions
|-- Partition 0
`-- Partition 1
    |
    v
    Flink Source
    |
    v
    parallel subtasks
```

5. Test Kafka Trước Khi Chạy Flink

Mục tiêu: xác nhận producer gửi được message, Kafka nhận được, consumer đọc được.

Mở 2 cửa sổ PowerShell.

Terminal 1: chạy consumer:

```
docker exec -it kafka /opt/kafka/bin/kafka-console-consumer.sh --topic
transactions --bootstrap-server localhost:9092
```

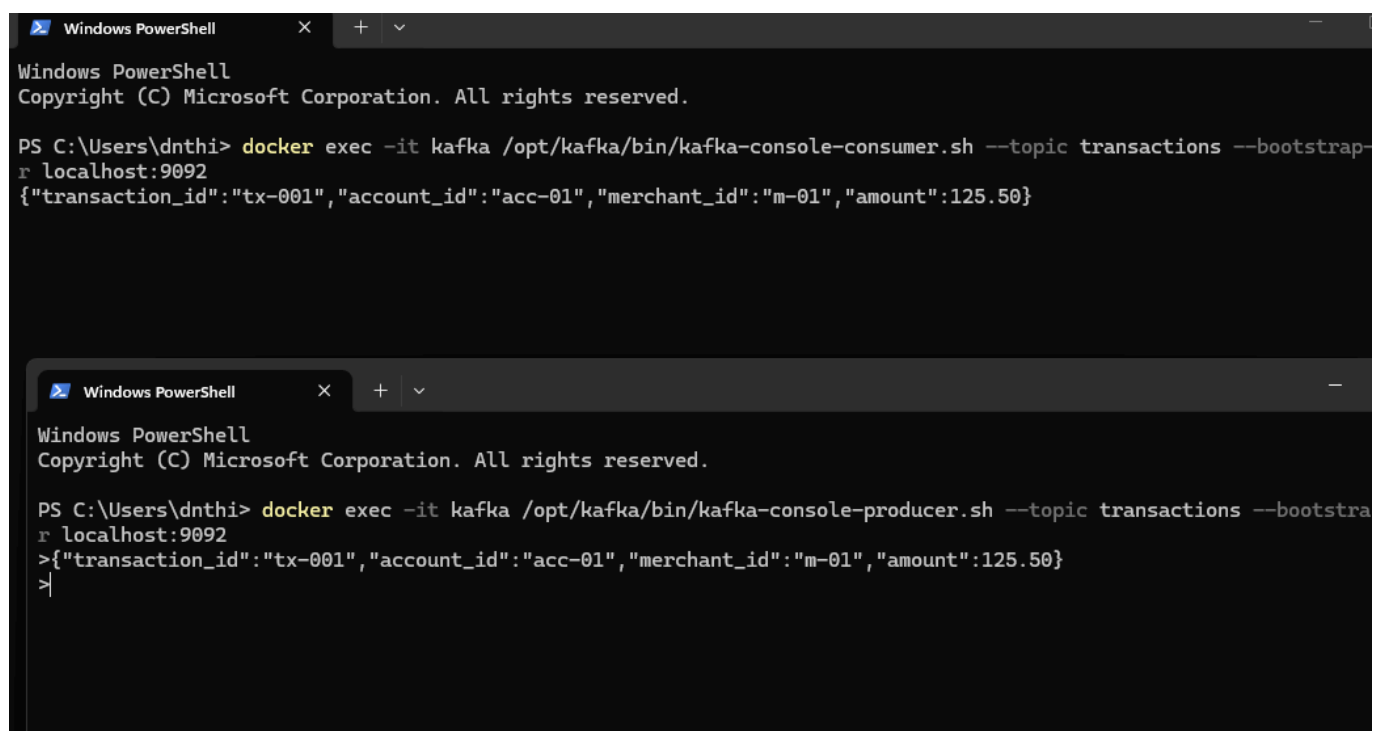
Terminal 2: chạy producer:

```
docker exec -it kafka /opt/kafka/bin/kafka-console-producer.sh --topic transactions --bootstrap-server localhost:9092
```

Nhập thử một event:

```
{"transaction_id":"tx-001","account_id":"acc-01","merchant_id":"m-01","amount":125.50,"event_time":"2026-08-14T10:00:01Z"}
```

Nếu terminal consumer hiện đúng JSON thì Kafka hoạt động ổn.



The image shows two screenshots of a Windows PowerShell terminal. The top screenshot shows the command to run the Kafka console consumer, which outputs a JSON message. The bottom screenshot shows the command to run the Kafka console producer, which outputs the same JSON message.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\dnthi> docker exec -it kafka /opt/kafka/bin/kafka-console-consumer.sh --topic transactions --bootstrap-server localhost:9092
{"transaction_id":"tx-001","account_id":"acc-01","merchant_id":"m-01","amount":125.50}
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\dnthi> docker exec -it kafka /opt/kafka/bin/kafka-console-producer.sh --topic transactions --bootstrap-server localhost:9092
>{"transaction_id":"tx-001","account_id":"acc-01","merchant_id":"m-01","amount":125.50}
>
```

6. Build Flink Job

Project dùng Maven. File [pom.xml](#) có các dependency chính:

Dependency	Mục đích
<code>flink-streaming-java</code>	DataStream API, <code>keyBy</code> , window, process.
<code>flink-connector-kafka</code>	Kafka source cho Flink.
<code>jackson-databind</code>	Parse JSON thành <code>Transaction</code> .
<code>maven-shade-plugin</code>	Đóng gói JAR để submit lên Flink.

Build:

```
mvn clean package
```

Kỳ vọng:

```
BUILD SUCCESS
```

JAR sau khi build:

```
dir target\*.jar
```

Ví dụ:

```
flink-transaction-demo-1.0-SNAPSHOT.jar
```

```
[WARNING] single version of the file is copied to the uber jar.
[WARNING] Usually this is not harmful and you can skip these warnings,
[WARNING] otherwise try to manually exclude artifacts based on
[WARNING] mvn dependency:tree -Ddetail=true and the above output.
[WARNING] See https://maven.apache.org/plugins/maven-shade-plugin/
[INFO] Replacing original artifact with shaded artifact.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 6.810 s
[INFO] Finished at: 2026-08-15T00:46:38+07:00
[INFO] -----
PS C:\Users\dnthi\OneDrive\Documents\BigData\Seminar\Theory\demo\flink-transaction-demo> |
```

7. Submit Flink Job

Copy JAR vào JobManager container:

```
docker exec flink-jobmanager mkdir -p /opt/flink/usrlib
docker cp target\flink-transaction-demo-1.0-SNAPSHOT.jar flink-
jobmanager:/opt/flink/usrlib/demo.jar
```

Kiểm tra:

```
docker exec flink-jobmanager ls -lh /opt/flink/usrlib/
```

Submit job:

```
docker exec flink-jobmanager flink run -d /opt/flink/usrlib/demo.jar
```

Nếu thành công:

```
Job has been submitted with JobID xxxxxxxxxxxxxxxxxxxx
```

Quay lại Flink Web UI:

```
http://localhost:8081
```

Kỳ vọng:

```
Running Jobs: 1  
Real-time Transaction Analytics  
Status: RUNNING
```


Apache Flink Dashboard

Version: 2.1.3 | Commit: 6cda56b @ 2026-06-08T04:56:20+02:00 | Message: 0

Real-time Transaction Analytics

[Cancel Job](#)

Job ID	46918fac9eaf88aeb86011f2377f2c9	Job State	RUNNING 2
Actions	Job Manager Log	Job Type	STREAMING
Start Time	2026-08-15 00:52:33.925	Duration	59s 359ms

[Overview](#) | [Exceptions](#) | [Data Skew](#) | [TimeLine](#) | [Checkpoints](#) | [Configuration](#)

Task Graph:

```

graph LR
    A["Source: Kafka Transactions Source -> Parse JSON -> Filter Invalid Transactions  
Parallelism: 2  
Backpressured (max): 0%  
Busy (max): 1%  
Data Skew: 0%"] -- HASH --> B["10s Transaction Analytics -> Sink: Realtime Analytics Output  
Parallelism: 2  
Backpressured (max): 0%  
Busy (max): 0%  
Data Skew: 0%"]
  
```

Name	Status	Bytes Read	Tasks
Source: Kafka Transactions Source -> Parse JSON -> Filter Invalid Transactions	RUNNING	0 B	2
10s Transaction Analytics -> Sink: Realtime Analytics Output	RUNNING	928 B	2

Trên Web UI cần quay được:

- Job name: **Real-time Transaction Analytics**.
- Job type: **STREAMING**.
- Parallelism: **2**.
- Job graph: Kafka Source -> Parse/Filter -> HASH -> Analytics -> Sink.
- Tab Checkpoints.

Lưu ý: Flink có thể **operator chaining**, nên Web UI có thể gộp nhiều operator thành vertex lớn. Đây là hành vi bình thường, không phải mất operator.

8. Chạy Transaction Producer

Mở PowerShell mới tại project:

```
cd "C:\Users\dnthi\OneDrive\Documents\BigData\Seminar\Theory\demo\flink-transaction-demo"
```

Chạy producer:

```
mvn exec:java "-Dexec.mainClass=com.seminar.flink.TransactionProducer"
```

Kỳ vọng terminal hiển thị liên tục:

```
=== Transaction Producer Started ===  
Sending transactions to Kafka topic: transactions  
  
SENT | acc-01 | m-02 | $325.00  
SENT | acc-03 | m-01 | $782.00  
SENT | acc-01 | m-03 | $149.00  
...
```

```
resolver-api-1.9.24.jar (157 kB at 696 kB/s)  
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".  
SLF4J: Defaulting to no-operation (NOP) logger implementation  
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.  
=== Transaction Producer Started ===  
Sending transactions to Kafka topic: transactions  
  
SENT | acc-01 | m-01 | $152.00  
SENT | acc-03 | m-03 | $102.00  
SENT | acc-02 | m-02 | $991.00  
SENT | acc-02 | m-02 | $227.00  
SENT | acc-02 | m-03 | $526.00  
SENT | acc-01 | m-02 | $404.00  
SENT | acc-03 | m-03 | $254.00  
SENT | acc-03 | m-02 | $978.00  
SENT | acc-03 | m-01 | $610.00  
SENT | acc-02 | m-02 | $953.00  
SENT | acc-02 | m-01 | $115.00  
SENT | acc-02 | m-03 | $635.00  
|
```

9. Xem Output Của Flink

Producer cứ để chạy. Mở PowerShell khác:

```
docker logs flink-taskmanager --tail 50
```

Muốn xem realtime:

```
docker logs -f flink-taskmanager
```

Kỳ vọng output dạng:

```
WINDOW | acc-01 | count=3 | total=$1556.00 | avg=$518.67
WINDOW | acc-02 | count=4 | total=$2301.00 | avg=$575.25 | ALERT: HIGH TRANSACTION VOLUME
WINDOW | acc-03 | count=3 | total=$966.00 | avg=$322.00
```

Ý nghĩa:

- **count**: số giao dịch trong window.
- **total**: tổng tiền trong window.
- **avg**: giá trị trung bình.
- **ALERT**: xuất hiện khi tổng tiền vượt ngưỡng.

```
PS C:\Users\dnthi> docker logs flink-taskmanager --tail 50
Caused by: org.apache.kafka.common.errors.CoordinatorNotAvailableException: The coordinator is not available.
2> WINDOW | acc-02 | count=2 | total=$1154.00 | avg=$577.00
1> WINDOW | acc-03 | count=3 | total=$862.00 | avg=$287.33
2> WINDOW | acc-01 | count=5 | total=$3186.00 | avg=$637.20 | ALERT: HIGH TRANSACTION VOLUME
2026-08-14 18:07:34,233 INFO org.apache.kafka.clients.consumer.internals.ConsumerCoordinator [] - [Consumer client
flink-transaction-analytics-1, groupId=flink-transaction-analytics] Discovered group coordinator kafka:29092 (id: 214
46 rack: null)
1> WINDOW | acc-03 | count=1 | total=$729.00 | avg=$729.00
2> WINDOW | acc-01 | count=5 | total=$3231.00 | avg=$646.20 | ALERT: HIGH TRANSACTION VOLUME
2> WINDOW | acc-02 | count=4 | total=$1785.00 | avg=$446.25
1> WINDOW | acc-03 | count=4 | total=$3255.00 | avg=$813.75 | ALERT: HIGH TRANSACTION VOLUME
2> WINDOW | acc-02 | count=3 | total=$1360.00 | avg=$453.33
2> WINDOW | acc-01 | count=3 | total=$892.00 | avg=$297.33
2> WINDOW | acc-02 | count=2 | total=$621.00 | avg=$310.50
1> WINDOW | acc-03 | count=6 | total=$2321.00 | avg=$386.83 | ALERT: HIGH TRANSACTION VOLUME
2> WINDOW | acc-01 | count=2 | total=$911.00 | avg=$455.50
2> WINDOW | acc-02 | count=3 | total=$1275.00 | avg=$425.00
```

10. Kiểm Tra Checkpoints

Trong Flink Web UI:

Job -> Checkpoints

Vì job có:

```
env.enableCheckpointing(10_000);
```

nên sau một lúc cần thấy checkpoint trạng thái **COMPLETED**.

Đây là phần cần show trong video để chứng minh Flink có cơ chế fault tolerance.

11. Thứ Tự Quay Video Demo

Nên quay theo thứ tự này để video gọn trong 4-5 phút:

1. Mở slide demo overview, nói workflow Kafka -> Flink -> realtime output.
2. Mở Flink Web UI, chỉ job đang **RUNNING**.
3. Chỉ Job Graph và giải thích ngắn: Source -> operators -> Sink.
4. Chỉ parallelism/subtasks nếu màn hình dễ nhìn.
5. Chạy hoặc show terminal **TransactionProducer**.
6. Show output trong **docker logs -f flink-taskmanager**.
7. Mở tab Checkpoints, chỉ checkpoint **COMPLETED**.
8. Chốt: demo đã thể hiện stream processing, dataflow, parallel execution, state/window và checkpoint.

12. Checklist Trước Khi Quay

- ☐ Docker Desktop đang chạy.
- ☐ **docker compose up -d** đã chạy thành công.
- ☐ Kafka topic **transactions** đã được tạo.
- ☐ **mvn clean package** thành công.
- ☐ Flink job submit thành công và đang **RUNNING**.
- ☐ Producer sinh transaction liên tục.
- ☐ Output window aggregate hoặc alert xuất hiện.
- ☐ Checkpoint có trạng thái **COMPLETED**.
- ☐ Terminal font đủ lớn, khoảng 16-20px.
- ☐ Browser zoom 110-125% nếu chữ nhỏ.
- ☐ Tắt thông báo hệ thống trước khi quay.

13. Cách Chạy Lại Vào Ngày Quay

Mai mở máy thì chạy theo thứ tự này.

13.1 Mở Docker Desktop

Mở Docker Desktop và đợi Docker Engine chạy ổn.

13.2 Mở PowerShell tại project

```
cd "C:\Users\dnthi\OneDrive\Documents\BigData\Seminar\Theory\demo\flink-transaction-demo"
```

13.3 Khởi động Kafka + Flink

```
docker compose up -d
```

Kiểm tra container:

```
docker ps
```

Phải có:

```
kafka  
flink-jobmanager  
flink-taskmanager
```

13.4 Kiểm tra Kafka topic

```
docker exec kafka /opt/kafka/bin/kafka-topics.sh --list --bootstrap-server  
localhost:9092
```

Nếu thấy:

```
transactions
```

thì OK.

Nếu không thấy thì tạo lại:

```
docker exec kafka /opt/kafka/bin/kafka-topics.sh --create --topic transactions --  
bootstrap-server localhost:9092 --partitions 2 --replication-factor 1
```

13.5 Submit Flink job lại

Vì `docker compose stop` hoặc restart thường giữ container, `demo.jar` có thể vẫn còn. Kiểm tra:

```
docker exec flink-jobmanager ls -lh /opt/flink/usrlib/demo.jar
```

Nếu có JAR thì submit job:

```
docker exec flink-jobmanager flink run -d /opt/flink/usrlib/demo.jar
```

Nếu mất JAR thì copy lại rồi submit:

```
docker exec flink-jobmanager mkdir -p /opt/flink/usrlib
docker cp target\flink-transaction-demo-1.0-SNAPSHOT.jar flink-
jobmanager:/opt/flink/usrlib/demo.jar
docker exec flink-jobmanager flink run -d /opt/flink/usrlib/demo.jar
```

13.6 Chạy producer ở PowerShell khác

Mở PowerShell mới:

```
cd "C:\Users\dnthi\OneDrive\Documents\BigData\Seminar\Theory\demo\flink-
transaction-demo"
mvn exec:java "-Dexec.mainClass=com.seminar.flink.TransactionProducer"
```

Sau đó mở Flink UI:

```
http://localhost:8081
```

Kỳ vọng luồng chạy lại:

```
Producer
|
v
Kafka
|
v
Flink RUNNING
|
v
Window / Alert
```

Kiểm tra output:

```
docker logs flink-taskmanager --tail 30
```

Hoặc xem realtime:

```
docker logs -f flink-taskmanager
```

14. Troubleshooting Nhanh

Không vào được Flink Web UI

Kiểm tra container:

```
docker ps
docker logs flink-jobmanager --tail 100
```

Kafka topic đã tồn tại

Nếu tạo topic báo topic đã tồn tại thì bỏ qua và kiểm tra bằng:

```
docker exec kafka /opt/kafka/bin/kafka-topics.sh --describe --topic transactions -
-bootstrap-server localhost:9092
```

Flink job không đọc được Kafka

Kiểm tra trong code Flink job:

```
.setBootstrapServers("kafka:29092")
```

Producer chạy từ máy host thì dùng:

```
bootstrap.servers = localhost:9092
```

Không thấy output ngay

Đợi ít nhất 10 giây vì job dùng tumbling window 10 giây.

Không thấy checkpoint completed

Đợi thêm 20-30 giây, sau đó mở tab Checkpoints trong Web UI. Nếu vẫn không có, kiểm tra job có đang **RUNNING** không.

15. Dọn Môi Trường Sau Demo

Dừng container:

```
docker compose down
```

Nếu muốn xóa cả dữ liệu container/network cũ:

```
docker compose down -v
```